



Valentin Legras

Principal C&SP Methodology Lead

Basel, Switzerland

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Profile

Principal C&SP Methodology Lead profile with 7+ years in regulated clinical data science across Roche, Novartis, and Sanofi. Expert in SAS and R, currently working about 40% in SAS and 60% in R, with hands-on experience across PK/PD, time-to-event, adverse events, laboratory, efficacy, dose, exposure, and safety data. Founded and led aNCA, an open-source R package adopted by 120+ scientists at Roche, delivering more than 2M CHF in annual savings. Combines CDISC (SDTM/ADaM), Pinnacle 21 checks, GxP, data privacy, anonymisation, Agile product ownership, and AI-assisted automation to build regulatory-grade tools from Phase 1 to market.

Work Experience

Senior Data Scientist

2020 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Engineered end-to-end clinical and statistical programming workflows, from data acquisition and preparation to analysis, visualization, reporting, and regulatory submission support across Phase 1 to 3 trials.
- Worked across broad clinical domains including PK/PD, time-to-event, adverse events, laboratory, efficacy, dose, exposure, and safety data, not only pharmacometric datasets.
- Developed SAS and R pipelines, R/Shiny applications, and Spotfire tools for real-time clinical data review, enabling faster cross-functional decisions on complex study data.
- Led the creation and QC of analysis-ready NCA, PK/PD, SDTM, and ADaM-compliant datasets from raw clinical data for statistical, pharmacometric, R, Phoenix, and NONMEM workflows, including Pinnacle 21 checks where applicable.
- Authored responses to health authority queries (FDA, EMA) and managed the QC and preparation of datasets and reports for electronic regulatory submissions, applying confidentiality, data privacy, and anonymisation principles when handling clinical data.
- Led a process improvement initiative: automation of ADPC, ADNCA, and ADPP datasets by converting legacy SAS programs into reproducible R pipelines, reducing manual effort by 90%. This new process is now used by 30 data scientists, saving them 70% of their time on these tasks.
- Scaled multi-agent AI coding workflows across the team, including AI-augmented issue creation, automated problem solving, and code review, boosting development velocity while upholding high engineering standards.
- Partnered with clinical, statistical programming, biostatistics, data management, and pharmacometric teams to ensure data integrity and deliver custom tools that streamlined analysis, modeling, simulation, and review.

aNCA Product Owner

2023 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Directed the end-to-end product lifecycle of the 'aNCA' open-source R package under Agile/Scrum, driving it from concept to enterprise-wide adoption by 120+ clinical and pre-clinical scientists at Roche.
- Led, coached, and mentored a core team of 7 developers, onboarded new clinical and pre-clinical colleagues, and managed a 15+ stakeholder network to define a product roadmap now supporting 2,000+ studies annually.
- Delivered more than 2M CHF in annual savings by transforming a manual, weeks-long workflow into a 90% automated process. This impact combines 5 FTEs and retired commercial software licenses.
- Established and enforced GxP-compliant software development life cycle practices for a regulated R package, including requirements translation, testing, documentation, code review, validation mindset, and mentorship.
- Contributed to the [pharmaverse](#) ecosystem by developing standardized modules for automated NCA, PK Table, Listing, and Graph (TLG) generation, using Git, CI/CD, Coding Agents (LLMs), and an AI interpretation bot in Python.

IT/Data Science Developer

Mar 2020 to Aug 2020

F. Hoffmann-La Roche, Basel, Switzerland

- Translated pharma scientists' requirements into technical specifications and IT/Data science tasks.
- Deployed several tools with R Shiny (data visualization) to support process modelers and formulators (incl. tidyverse packages).
- Used Git, Jenkins, and Google Cloud Platform to build, test, and deploy applications.
- Automated database updates on Google Spreadsheet with R (incl. VBA coding), data management and data merging.
- Automated PowerPoint reports with R (officer package) for daily reports.

Data Scientist 
Cofidis, Lille, France

Oct 2019 to Feb 2020

- Analyzed customer loan data in R and built machine learning models (Logistic Regression, Gradient Boosting, Random Forest) to predict contract default.
- Recommended data collection strategies to optimize the rate of returned and completed contracts.

Biostatistical Application Developer



May 2019 to Aug 2019

Novartis AG, Basel, Switzerland

- Built an R Shiny application to visualize clinical data, tables, figures, and listings interactively during DMC (Data Monitoring Committee) reviews, then generalized the application across studies.
- Created dynamic HTML reports with R Markdown (dplyr, ggplot, plotly, DT) using CDISC SDTM and ADaM datasets.

SAS Developer



Jun 2018 to Aug 2018

Sanofi R&D, Paris, France

- Developed and validated SAS (BASE/MACRO/SQL) programs for TLG reporting and built a data-mapping tool generating XML for an AI project, using SDTM/ADaM specifications and clinical programming standards.

SAS and IT Developer



Apr 2017 to Aug 2017

Sanofi R&D, Paris, France

- Migrated and optimized SAS programs for clinical trial data in a new statistical environment, enabling biostatisticians to run their analyses.

Skills

Clinical & Statistical Programming

SAS (BASE/MACRO/SQL), SAS macros, R, TLG automation, clinical data review, CDISC (SDTM/ADaM), Pinnacle 21, GxP

Clinical Data Domains

PK/PD, NCA, time-to-event, adverse events, laboratory, efficacy, dose, exposure, safety, data anonymisation

AI & Automation

Claude (Sonnet/Opus), coding agents (Ona, Claude Code), multi-agent workflows, LLM-assisted development, prompt engineering, AI-driven automation pipelines

Software & Automation

R package development, Shiny, Python, SQL, Git, CI/CD, Jenkins, Google Cloud Platform, Agile/Scrum, SDLC, testing, documentation

Privacy & Compliance

data privacy, confidentiality levels, controlled access, anonymisation, Good Clinical Practice (GCP), regulatory submissions, FDA/EMA queries

Visualization & Reporting

R Shiny, ggplot, R Markdown, Spotfire/Tableau, tables, listings, figures

Leadership & Delivery

team leadership, mentoring, product ownership, stakeholder management, cross-functional collaboration

Languages

English (Fluent) ●●●●●
French (Native) ●●●●●
Spanish (B2) ●●●○○

Educational Background

Master's Degree, Software Engineering and Statistics

2017 to 2020

Polytech Lille, Villeneuve-d'Ascq, France

Completed alongside internships/apprenticeship at Sanofi, Novartis, Cofidis, and Roche.

Technical Degree, Statistics and Business Intelligence

2015 to 2017

Institut Universitaire de Technologie, Vannes, France

Presentations

Conference Presentations

- [Revolutionizing Data Exploration: An Interactive Journey through R Shiny](#). Legras V., PHUSE 2023.
- [R Scripts to Streamline Non-Compartmental Analysis \(NCA\)](#). Legras V., Suekuer E., PHUSE 2024 (aNCA open-source).
- [Sunburst Plots for Analysing the Impact of Drugs on Cytokine Release Syndrome \(CRS\) Management](#). Legras V., PAGE 2024.
- [Automated QC Framework for Reliable ADNCA Dataset Generation with admiral](#). Legras V., Aspridis L., PHUSE 2025.

Hobbies

Roller skating, Ice skating, Surfing, Basketball, Cycling, Drawing, Coding ([stock market analysis](#)).